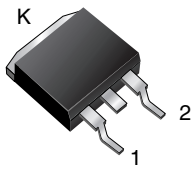
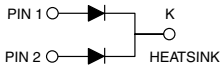
Dual Common Cathode Schottky Rectifier

High Barrier Technology for Improved High Temperature Performance

D²PAK (TO-263AB)



MBRB30HxxCT



FEATURES

- Power pack
- Guardring for overvoltage protection
- Lower power losses, high efficiency
- Low forward voltage drop
- Low leakage current
- High forward surge capability
- High frequency operation
- Meets MSL level 1, per J-STD-020, LF maximum peak of 245 °C
- AEC-Q101 qualified available
- Automotive ordering code: base P/NHM3
- Material categorization: for definitions of compliance please see www.vishay.com/doc?99912

RoHS COMPLIANT
HALOGEN FREE

TYPICAL APPLICATIONS

For use in low voltage, high frequency rectifier of switching mode power supplies, freewheeling diodes, DC/DC converters, or polarity protection application.

MECHANICAL DATA

Case: D²PAK (TO-263AB)

Molding compound meets UL 94 V-0 flammability rating Base P/N-M3 - RoHS-compliant, halogen-free, commercial grade

Base P/NHM3 - RoHS-compliant, halogen-free, AEC-Q101 qualified

Terminals: matte tin plated leads, solderable per J-STD-002 and JESD 22-B102

M3 suffix meets JESD 201 class 1A whisker test, HM3 suffix meets JESD 201 class 2 whisker test

Polarity: as marked

PRIMARY CHARACTERISTICS	
$I_{F(AV)}$	2 x 15 A
V_{RRM}	45 V, 60 V
I_{FSM}	150 A
V_F	0.56 V, 0.59 V
I_R	80 µA, 60 µA
T_J max.	175 °C
Package	D²PAK (TO-263AB)
Circuit configuration	Common cathode

MAXIMUM RATINGS (T_A = 25 °C unless otherwise noted)

PARAMETER		SYMBOL	MBRB30H45CT	MBRB30H60CT	UNIT
Maximum repetitive peak reverse voltage		V_{RRM}	45	60	V
Working peak reverse voltage		V_{RWM}	45	60	V
Maximum DC blocking voltage		V_{DC}	45	60	V
Maximum average forward rectified current (fig. 1)	total device	$I_{F(AV)}$	30		A
	per diode		15		
Peak forward surge current 8.3 ms single half sine-wave superimposed on rated load per diode		I_{FSM}	150		A
Peak repetitive reverse surge current per diode at t_p = 2 µs, 1 kHz		I_{RRM}	1.0	0.5	A
Peak non-repetitive reverse energy (8/20 µs waveform)		E_{RSM}	25	20	mJ
Non-repetitive avalanche energy per diode at 25 °C, I_{AS} = 4 A, L = 10 mH		E_{AS}	80		mJ
Electrostatic discharge capacitor voltage human body model: C = 100 pF, R = 1.5 kΩ		V_C	25		kV
Voltage rate of change (rated V_R)		dV/dt	10 000		V/µs
Operating junction and storage temperature range		T_J, T_{STG}	-65 to +175		°C

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ELECTRICAL CHARACTERISTICS (T_A = 25 °C unless otherwise noted)								
PARAMETER	**TEST CONDITIONS**		**SYMBOL**	**MBRB30H45CT**		**MBRB30H60CT**		**UNIT**
				TYP.	**MAX.**	**TYP.**	**MAX.**	
Maximum instantaneous forward voltage per diode	I_F = 15 A	T_C = 25 °C	V_F (1)	-	0.62	-	0.68	V
	I_F = 15 A	T_C = 125 °C		0.49	0.56	0.55	0.59	
	I_F = 30 A	T_C = 25 °C		-	0.73	-	0.83	
	I_F = 30 A	T_C = 125 °C		0.62	0.67	0.68	0.71	
Maximum reverse current per diode at working peak reverse voltage		T_J = 25 °C	I_R (2)	-	80	-	60	µA
		T_J = 125 °C		5.0	15	4.0	15	mA

Notes

(1) Pulse test: 300 µs pulse width, 1 % duty cycle

(2) Pulse test: pulse width ≤ 40 ms

THERMAL CHARACTERISTICS (T_A = 25 °C unless otherwise noted)				
PARAMETER	**SYMBOL**	**MBRB30H45CT**	**MBRB30H60CT**	**UNIT**
Typical thermal resistance junction to case per diode	$R_{\theta JC}$	1.5		°C/W

ORDERING INFORMATION (Example)					
PACKAGE	**PREFERRED P/N**	**UNIT WEIGHT (g)**	**PACKAGE CODE**	**BASE QUANTITY**	**DELIVERY MODE**
D^2PAK (TO-263AB)	MBRB30H45CT-M3/I	1.35	I	800/reel	Tape and reel
D^2PAK (TO-263AB)	MBRB30H45CTHM3/I (1)	1.35	I	800/reel	Tape and reel

Notes

(1) AEC-Q101 qualified

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RATINGS AND CHARACTERISTICS CURVES (T_A = 25 °C unless otherwise noted)

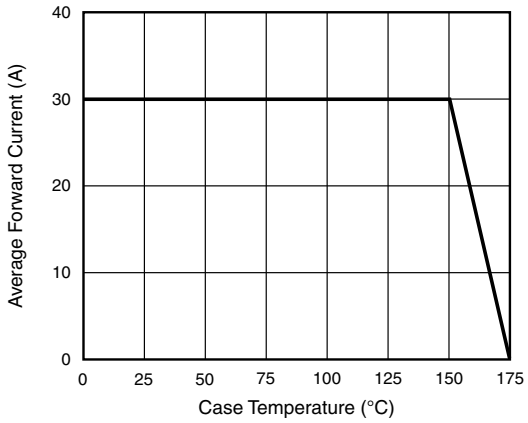


Fig. 1 - Forward Derating Curve

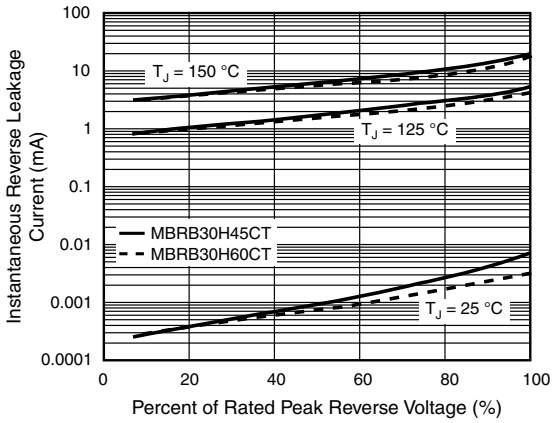


Fig. 4 - Typical Reverse Characteristics Per Diode

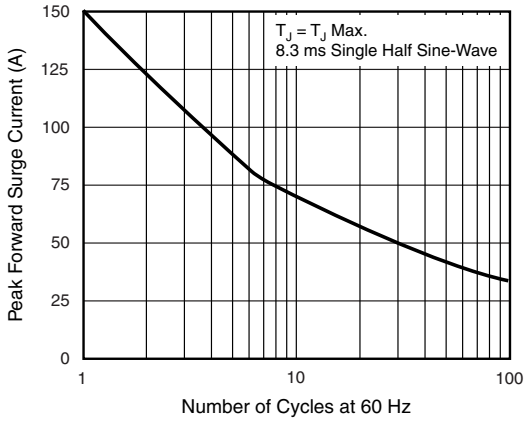


Fig. 2 - Maximum Non-Repetitive Peak Forward Surge Current Per Diode

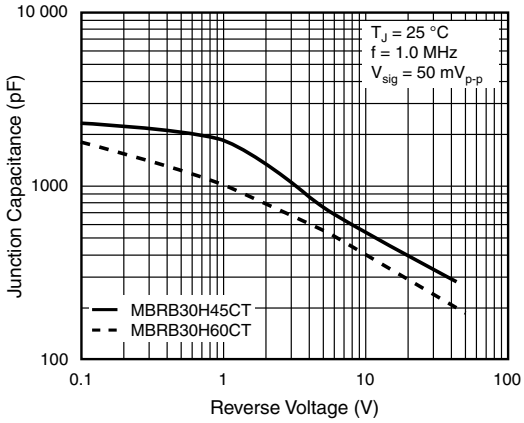


Fig. 5 - Typical Junction Capacitance Per Diode

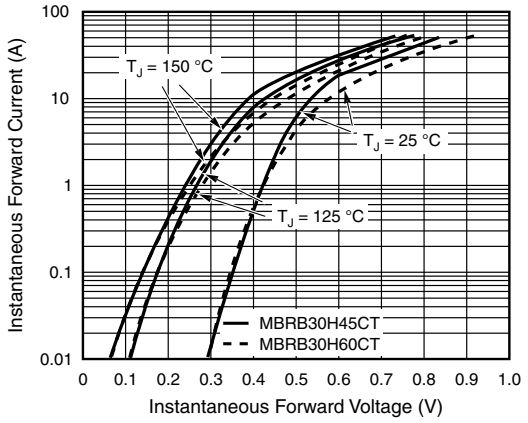


Fig. 3 - Typical Instantaneous Forward Characteristics Per Diode

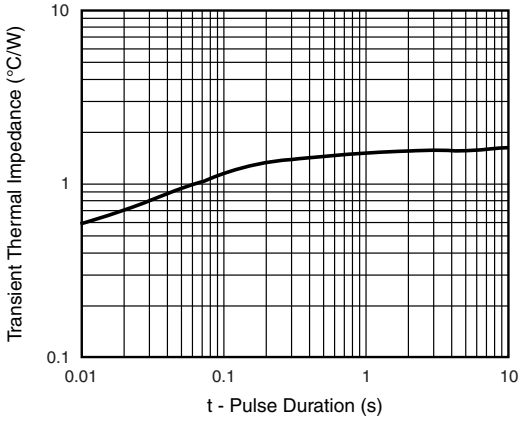
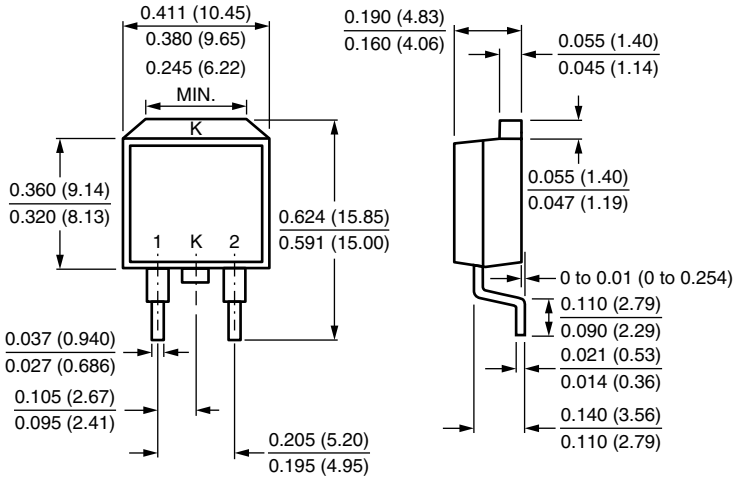


Fig. 6 - Typical Transient Thermal Impedance Per Diode

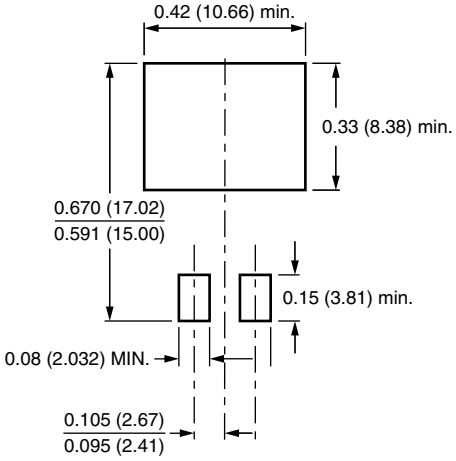
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PACKAGE OUTLINE DIMENSIONS in inches (millimeters)

D^2PAK (TO-263AB)



Mounting Pad Layout



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